

Indian Institute of Technology Dharwad

CS201L: Artificial Intelligence Laboratory

Practice Set: Distance-Based Classification using K-Nearest Neighbors (KNN)

Date: 02-02-2026

Problem Statement

You are given a real-world multiclass classification dataset as a CSV file named `Date_Fruit_Datasets.csv`, obtained from Kaggle. This dataset contains 898 samples with 34 numerical features extracted from date fruit images and 1 target class attribute.

The objective of this practice set is to implement the complete machine learning workflow using:

- Data loading
- Train-validation-test splitting
- Saving datasets as CSV files
- Model selection using validation data
- Distance-based classification using KNN
- Performance evaluation using standard classification metrics

Dataset Properties

- Total samples: 898
- Total features: 34 (numerical)
- Target attribute: `Class` (multiclass)
- Data format: CSV file

Tasks

Write a Python program to complete the following tasks.

Part A: Data Preparation and Splitting

1. Load the dataset: `df = pd.read_csv('Date_Fruit_Datasets.csv')`
2. Separate features (X) and target (y).
3. Split the dataset into:
 - Training set: 60%
 - Validation set: 20%
 - Test set: 20%

Use stratified sampling with `random_state = 42`.

4. Save the generated splits as CSV files:
 - `DateFruit_Train.csv`

- DateFruit_Validation.csv
- DateFruit_Test.csv

Note: The `Class` attribute must be included in all saved CSV files.

Part B: KNN Model Selection using Validation Data

1. Load training, validation, and test datasets from:

- DateFruit_Train.csv
- DateFruit_Validation.csv
- DateFruit_Test.csv

2. Apply KNN classification using the following values of K :

$$K = 1, 3, 7, 11$$

3. For each value of K , evaluate the model on the **validation data** and compute:

- Accuracy
- Precision (macro, micro)
- Recall (macro, micro)
- F1-score (macro, micro)

4. Plot **Accuracy vs. K** using validation accuracy.
5. Select the **best K** based on highest validation accuracy.

Part C: Final Evaluation on Test Data

1. Using the best value of K selected from validation:

- Evaluate the model on the **test data**

2. Report the following test metrics:

- Confusion Matrix
- Accuracy
- Precision (macro, micro)
- Recall (macro, micro)
- F1-score (macro, micro)

3. Provide a brief interpretation of the model performance based on the metrics.

Important Notes

1. For train-validation-test split (60%-20%-20%), use:

```
X_train, X_temp, y_train, y_temp = train_test_split(
    X, y, test_size=0.4, stratify=y, random_state=42)
X_val, X_test, y_val, y_test = train_test_split(
    X_temp, y_temp, test_size=0.5, stratify=y_temp,
    random_state=42)
```

Keep `random_state=42` for all splits to ensure consistency.

2. Import required libraries:

```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import (accuracy_score, precision_score,
                             recall_score, f1_score, confusion_matrix)
```

3. For multiclass metrics, use:

```
average='macro'
average='micro'
```

4. Required plot:

- Accuracy vs. K on validation data

5. Model selection rule:

- Use **validation data only** to select the best K
- Use **test data only once** for final evaluation

6. Reporting rules:

- Report validation metrics for all K values
- Report test metrics only for the best K

Performance Metrics

- **Accuracy:** Overall correctness
- **Precision (Micro):** Global calculation across all classes
- **Precision (Macro):** Unweighted mean per class
- **Recall (Micro):** Global calculation across all classes
- **Recall (Macro):** Unweighted mean per class
- **F1-score (Micro):** Harmonic mean of micro precision and recall
- **F1-score (Macro):** Unweighted mean per class

— End of Practice Set —